DESIGN OF A LINEAR SERVO ACTUATOR

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I. PROBLEM STATEMENT

A linear actuator is a device that creates linear motion in a straight line, usually achieving this by converting circular motion to linear motion[5]. There are many types of linear actuators, but the one we will be focusing on is called a mechanical linear actuator [1]. It consists of a rack, pinion [4], and motor where, given the pinion rotates a certain amount, the rack moves a certain linear distance [4]. Our group set out to design a linear actuator that has a linear distance of at least 20 millimeters in one rotation of its motor. In addition, the motor and the pinion must be concealed inside a mount. The pinion, rack, and mount will be 3D printed using a Prusa 3D Printer, and we must use as little material as possible. A typical linear actuator costs around $40[2]; therefore, a successful design will cost less than that. The cost of the printed materials will be determined by how much material is used and how long it takes to print. The rest of the materials used have been provided, and their costs will be taken into account as well.

II. CONCEPT

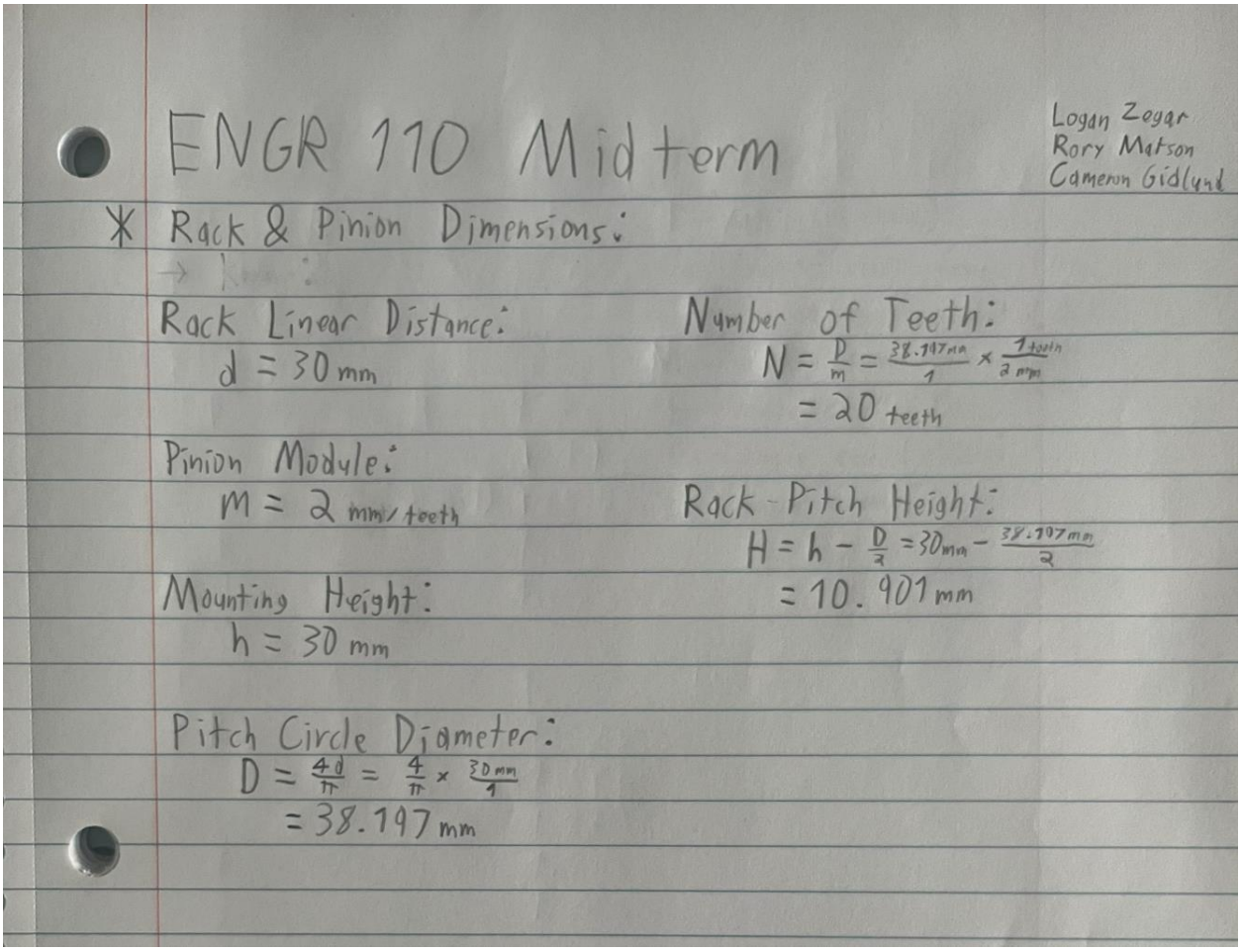


Figure 1: Equations used to design our rack and pinion

We started by determining the dimensions of the pinion. Since the actuator must have a linear distance of 20 mm, we designed ours to have a distance of 30 mm and a module of 2. Using that information and the equations[4] provided by Dr. Alam, we determined the pitch circle diameter and the number of teeth. As for the diameter of the inner circle of the pinion, we made it equal to the diameter of the motor we are using. We then found the dimensions of the rack, starting by designing a mounting height of 30 mm and a total length of 100 mm. Using the

mounting height, pitch circle diameter, and the equation[4] provided by Dr. Alam, we determined the rack pitch height.

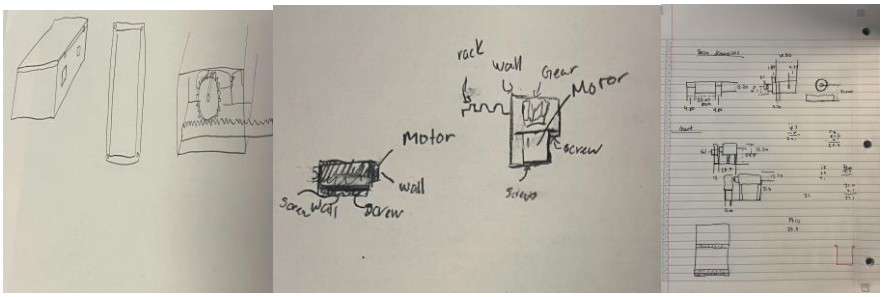


Figure 2-4: Our initial concept ideas for the rack, pinion, and mount

Once we had all the dimensions we needed, we began drafting a concept, including the mount that would be used to hold everything together. Our first concept consisted of a box enclosing the pinion and motor, with the rack being moved through a hole. However, we decided that approach would use too much printing material, so we created a design that only encloses the motor, leaving the pinion and rack open. This design makes it easier to access the rack and pinion, while still holding everything together.

Once we had determined what we were going to make, we made a final concept of the linear actuator. One major difference between our initial and final concept was the contact point between the pinion and motor. We found that while the pinion fits into the motor, it would not rotate when the motor rotated. This is because it is not tight enough to create rotation. To fix this, we added a small sheet of paper wedged between the pinion and the motor.

III. DESIGN

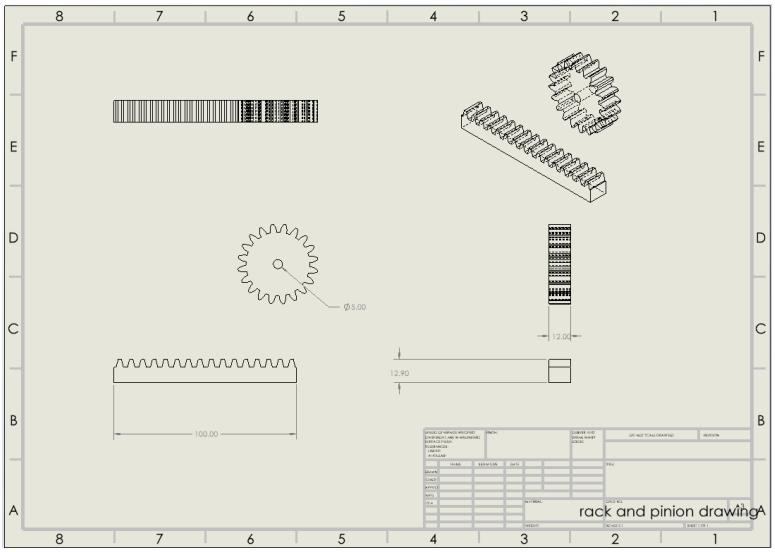


Figure 6: Model and drawing of the rack & pinion

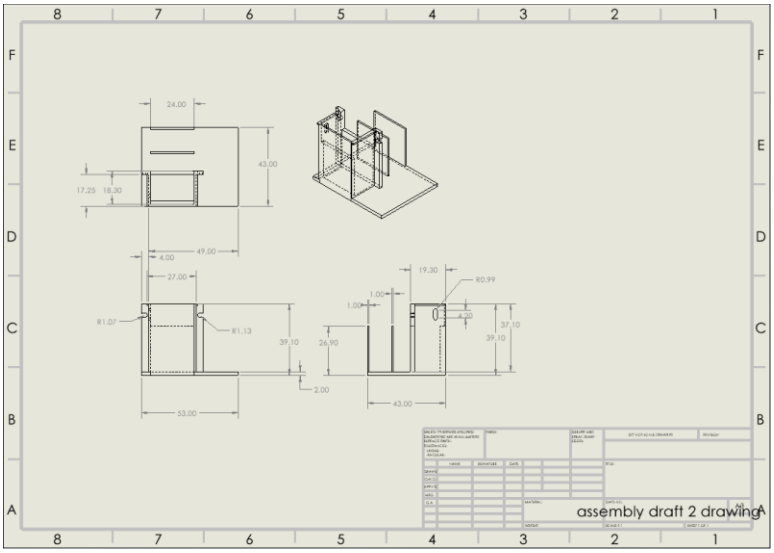


Figure 7: Model & drawing of the mount

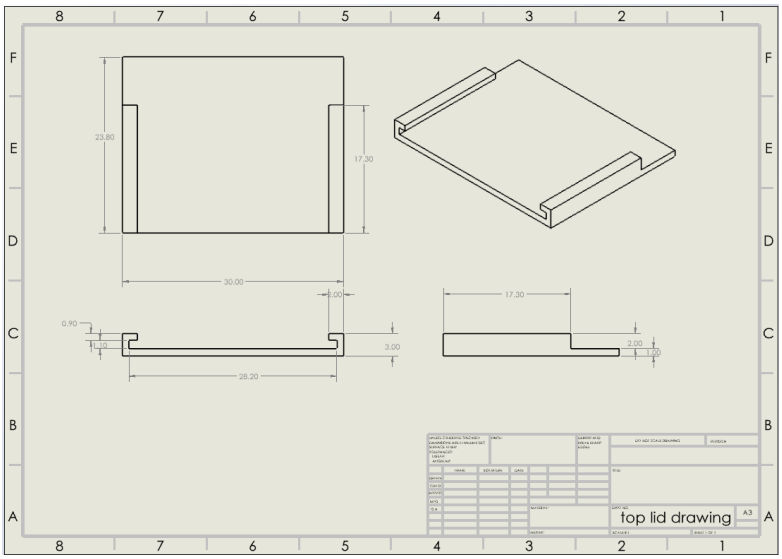


Figure 8: Model and drawing of the mount lid

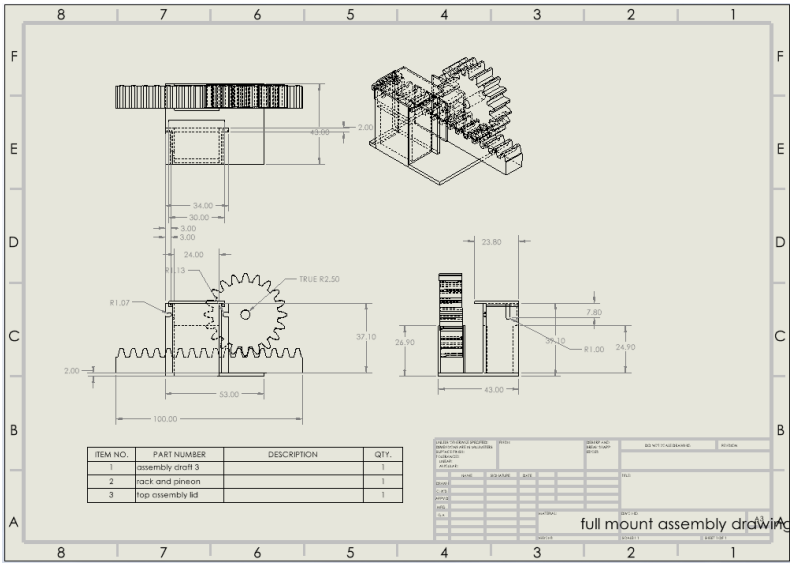


Figure 9: Assembly of every part made using SolidWorks (including motor for reference)

IV. BILL OF MATERIALS

Linear Actuator Material List			
Number	Item Name	Quantity	Source
1	MG90S Geared Micro Servo Motor	1	Amazon Link
2	Digital Multi Servo Speed Controller	1	Amazon Link
3	Battery Holder	1	Provided By Dr. Alam
4	AA Batteries	1	Amazon Link
6	M2 screws and nuts	1	Amazon Link
7	Pinion	1	3D Printed Using Prusa 3D Printer
8	Rack	1	3D Printed Using Prusa 3D Printer
9	Mount	1	3D Printed Using Prusa 3D Printer

Figure 10: List of materials used in our design with available sources

V. COST ANALYSIS

When comparing the cost of 3d printing and just buying a linear actuator, 3d printing is cheaper by about $30. When you break down every material needed by individual piece's price, it cuts the cost down significantly. None of the prices are calculated in shipping, but that is going to be something to consider, as seen in figure 10. While a 3D printer is a costly thing to consider, it is worth it, with how much can be saved over time. If a 3D printer is out of budget, most public libraries have 3D printers, but they have a limit on the amount of filament used.

Amazon Price	**$40**					**Total Cost**	**$9.86**
Basic Materials		**3D Printed Materials**					
Item	Cost	Given Information	Cost		Amount	Cost	
Batteries (4)	$2.24	Time Cost	$1 / hour	Time	2.5 hours	$2.50	
Motor (1)	$2.25	Material Cost	$20 / kg	Material	0.06 kg	$1.20	
Servo (1)	$1.67			**Total**		**$3.70**	
Total	**$6.16**						

Figure 11: A breakdown of the costs of our design [2]

VI. RESULTS



Figure 12-13: Test of the linear actuator, displaying the total distance traveled by the linear actuator

Our final design covers a linear distance of 30 mm in 1 full motor rotation, the exact same as what we designed the linear actuator to do. This makes the design a very accurate system to use. The main source of error in this design comes from the friction between all points of contact: the motor and pinion, the pinion and rack, and the rack and ground. However, as demonstrated by figure 13, this source or error is negligible.

VII. DISCUSSIONS

This midterm project proved difficult and time-consuming. However, the cost difference of 3D printing proves to be the best option on a budget. There are many ways to access 3D printers without buying them such as the library. When it comes to concern over getting the math correct, the equations needed for a rack and pinion[4] are available online. If you have access to the target measurements for the calculations, math should not be a big concern.

The most difficult part of the whole project was getting the design correct. Creating the correct design that utilizes the least number of materials was difficult. Since the print needs to be exact,

there were several moments where minor adjustments were made. This proved to be time-consuming and frustrating.

ACKNOWLEDGEMENTS

1. SolidWorks 3D Software
2. Prusa 3D Printer & Software
3. Dr. Md Erfanual Alam

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[1]Actuonix Motion Devices. (2017). *Linear Actuators: What Are They, How Do They Work?* Www.actuonix.com. https://www.actuonix.com/what-are-linear-actuators

[2]Amazon. (2024). *ECO-WORTHY Heavy Duty 330lbs/1500N Solar Tracker Linear Actuator Multi-Functions with Mounting Brackets (12V, 12") IP54 Waterproof 300mm Stroke Linear Motion Actuator*. Amazon. https://www.amazon.com/product-reviews/B00NM8H5SW/ref=cm_cr_getr_d_paging_btm_next_10?ie=UTF8&reviewerType=all_reviews&returnFromLogin=1&sortBy=recent&pageNumber=10

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